



Influence of malt source on beer chemistry, flavor, and flavor stability

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ABSTRACT

Beverage quality in the brewing industry is heavily influenced by ingredient properties. The contribution of raw ingredients such as yeast and hops to beer flavor is well understood. However, the influence of barley genotype and/or environment on flavor (the malt 'source') is largely unexplored. Here, a study was performed to determine (i) if there are metabolite differences among six commercial malt sources, (ii) if differences in malt chemistry are reflected in the chemistry of the beer, and (iii) if the differences in the beer chemistry impact sensory attributes of beer, through flavor and flavor stability. Six distinct sources of malts (six varieties from three maltsters) were brewed into six beers using a recipe designed to evaluate differences in flavor. Metabolomics and ionomics was used to characterize chemical variation among the six malts and beers using UHPLC- and HILIC-MS (non-volatile metabolites), HS-SPME/GC-MS (beer volatiles), and ICP-MS (malt metals). These analyses detected a total of 5042 compounds in malt, of which 217 were annotated and included amines, amino acids, fatty acids/lipids/fatty acyls, saccharides/glucosides/sugar acids/sugar alcohols, carboxylic acid derivatives, organic acids, phenolics/benzenoids, purines, pyrimidines/pyridines, terpenes, and organosulfurs. A total of 4568 compounds were detected in beer, of which 246 were annotated and included esters, aldehydes, and alcohols. Statistical analysis revealed chemical variation among the six malts (50/217 malt metabolites varied) and beers (150/246). The six beers were evaluated for flavor using a modified descriptive analysis for 45 sensory traits at 0, 4, and 8 weeks of storage at 4 °C. Principal component analysis of the sensory data revealed flavor differences among the six beers at 8 weeks, and the malt-type Full Pint was described as *fruity* and Meredith as *corn chip*. The metabolite and sensory data were integrated and revealed associations between flavor profiles in beer and the annotated malt and beer. The *fruity* or *corn chip* flavor profiles in beer were associated with beer purines/pyrimidines, volatile ketones, amines, and phenolics, and malt lipids, saccharides, phenols, amines, and alkaloids. Taken together, these data support a role of malt source in beer flavor and flavor stability. As a raw ingredient, malting barley genotypes can be evaluated for a contribution to flavor, and this may be a future target for plant breeding, agronomy, and malting efforts to selectively improve flavor, flavor stability, and quality in beer.

1. Introduction

Understanding the mechanisms of how raw ingredients and their interactions influence beer quality is critical for the brewing industry (Iimure & Sato, 2013). Beer organoleptic traits are highly discernable by consumers based on flavor (taste and aroma), appearance, and

mouthfeel. These organoleptic attributes are variable and define the overall beer style and drive consumer trends (Briggs, 2004). Variation in the genetics of raw ingredients such as yeast (strain-type) and hop (cultivar) are critical to developing beer styles, such as Hefeweizen beer created with Hefeweizen Ale yeast (Gallone et al., 2016; White & Zainasheff, 2010), and Imperial IPA beer created using Cascade,

Abbreviations: ICP-MS, inductively coupled plasma mass spectrometry; UHPLC-MS, ultra high performance liquid chromatography mass spectrometry; HS-SPME/GC-MS, headspace solid-phase microextraction gas chromatography mass spectrometry; HILIC-MS, hydrophilic interaction chromatography mass spectrometry; QDA, quantitative descriptive analysis

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Table 1
Barley malt sources used in this study.

Barley genotype	Maltster	Malthouse location	Moisture (%)	FGDB (%)	Color (°)	DP (L)	AA (DU)	TP (%)	SP (%)	S/T	FAN (ppm)	Viscosity (CPU)	β-Glucan	Friability (%)	pH
Metcalfe	Malt-europ	Great Falls, MT	4.9	82.4	1.55	127	62.7	10.86	4.1	37.8	177	1.5	96	93	5.9
Expedition	Malt-europ	Great Falls, MT	5	81.1	1.74	144	58.7	12.24	4.82	39.4	204	1.48	105	89.5	5.9
Meredith	Rahr	Alix, AB	5.5	83.3	1.83	156	59.8	10.91	4.65	42.6	189	1.49	115	90.5	6.0
Full Pint	Briess	Chilton, WI	9	79.6	1.78	181	82	13.62	4.78	35.1	173	1.59	220	55.5	5.9
Polar Star	Cargill	Biggar, SK	4.9	82.3	2.04	156	65.7	10.98	4.64	42.2	173	1.48	72	93	6.1
Copeland	Rahr	Alix, AB	5.2	82.4	1.49	126	52.1	11.14	4.25	38.1	161	1.5	130	89.2	6.1

FGDB – fine-grind, dry basis; Color – based on Standard Reference Method (SRM); DP – diastatic power, based on Lintner units; AA – alpha amylase, based on diastatic units (DU), 30 or above is required for proper conversion; TP – total protein, target of < 14%; SP – soluble protein, based on dry basis; S/T – soluble/total ratio, a minimum of 30 is required to prevent lautering issues; FAN – free amino nitrogen, standard value is 180 ppm and above; Viscosity – typically 1.45–160 cP units (CPU); β-glucan – < 180 indicates good lautability, but this test only indicates the number of molecules found, not the molecular weight; Friability – indicator of lautering performance, > 90% indicates good friability, < 90% indicates under-modification (high viscosity polysaccharides such as β-glucan, leading to lautering difficulties).

Columbus, and Amarillo hops (Hieronymus, 2012). Further, beer flavor is largely impacted by its flavor stability. Beer is a chemically unstable product (Bokulich & Bamforth, 2013) and its composition continuously changes during storage (B. Vanderhaegen, Neven, Verachtert, & Derdelinckx, 2006; B.D. Vanderhaegen, Filip, Daenen, 2007).

The contribution of barley to beer flavor is largely developed through the malting process, as well as its macro-chemical contribution to brewing. As an ingredient, malt provides saccharides, proteins, free amino nitrogen (FAN), and enzymes that facilitate fermentation reactions in brewing, and the content of these traits is used to describe the ingredient's 'malt quality'. These malt quality traits can influence flavor, such as excess saccharides that are not completely fermented result in sweet beer (Bamforth, 1993; Bokulich & Bamforth, 2013; Fox, 2010), and excess FAN components that influence fermentation rates to result in higher levels of 2,3-butanedione and diacetyl that produce butterscotch/buttered popcorn flavor (Dong et al., 2015; Gupta, Abu-Ghannam, & Gallagher, 2010). Further, recent studies have characterized the metabolite composition of malt, notably small molecules (< 1200 Da), some of which are known to influence flavor (Heuberger et al., 2012; Heuberger et al., 2016; Heuberger, Broeckling, Kirkpatrick, & Prenni, 2014; Heuberger, Robison, Lyons, Broeckling, & Prenni, 2014; Hughey, McMinn, & Phung, 2015). During malting, the outcome (e.g. color and flavor) of malt is largely controlled by kilning at specific temperatures and lengths of time, for example differentiating pale and chocolate malt. The kilning reaction produces Maillard reaction products such as maltozazine, maltol, isomaltol, and ethyl maltol that provide bread, caramellic, or cotton candy like flavor to beer (Coghe, Martens, D'Hollander, Dirinck, & Delvaux, 2004). Malt alkaloids have also been characterized to influence beer flavor, specifically the horadates and beer astringency (Inui, Tsuchiya, Ishimaru, Oka, & Komura, 2013; Kageyama, Inui, Fukami, & Komura, 2011; Kreisz, 2009; O'Rourke et al., 2002). However, while the individual flavor attributes of these metabolites is known, the influence of variation in malt metabolites on beer flavor is not well understood.

It has been well demonstrated that barley malting quality is influenced by genetic, environmental factors, and their interactions (BA, 2014, 2016; Iimure & Sato, 2013; Li, Wang, & Liu, 2017; Schwarz & Li, 2010; Shewry, 2014; Turner et al., 2013). An example is row-type, for which 2-row genotypes have higher saccharide/protein ratios compared to 6-row genotypes. 6-row types are often brewed with adjunct grain to account for the reduced saccharides (e.g. corn, wheat, rice). Further, both genetics and environment affect the overall metabolite composition of the barley and subsequent malt, including saccharides, lipids, and amino acids (Heuberger, Broeckling, et al., 2014). Additional research has described the relationships between variation in malting quality traits and metabolites, specifically for β-glucans, diastatic power, α-amylase, and extract (BMBRI, 2017; Bokulich &

Bamforth, 2013; Briggs, 2004; D. Briggs, 1998; Cozzolino, Roumeliotis, & Eglinton, 2015; Gupta et al., 2010; Schwarz & Li, 2010).

In 2014, the Brewers Association, (BA, an organization that represents > 5000 United States breweries) published a white paper that outlined ideal barley malt standards and attributes: "Malting Barley Characteristics for Craft Brewers" (BA, 2014). In the white paper, it is stated that, "U.S. craft brewers have access to, and increasingly use, diverse malts from around the world in order to flavor differentiate their brands from all others." However, in the white paper, the BA also identified the challenge of discriminating flavor based on the source (genotype and environment). The BA white paper identifies "flavor" as a gap in our knowledge for malting and brewing: "there is poor understanding of the origins of flavor, whether genetic or arising during malting or kilning or from any combination of factors." In response to this, a study was performed which used a plant genetics approach and demonstrated a direct association between barley genotype and beer flavor, independent of malt quality traits (Herb et al., 2017). Together, these studies suggest that barley varieties contain metabolite differences that influence beer flavor.

Here, a study was performed to investigate the chemical influence of malt source (the combined variation due to genotype and environment, including maltster) on the chemical composition of the resulting beers, as well as effects on flavor and flavor stability. Six distinct sources of malts (six varieties from three maltsters from two countries) were brewed into six beers using a recipe designed to evaluate differences in flavor. Further, a comprehensive non-targeted metabolomics and ionomics method was used to evaluate chemical variation among the six malts. The specific objectives for this study were to understand (i) if there would be metabolite differences among six commercial barley sources, (ii) if differences in barley chemistry are reflected in the chemistry of the beer, and (iii) if the differences in the beer chemistry impact sensory attributes of beer, through flavor and flavor stability.

2. Materials and methods

2.1. Plant material

A total of six malt sources were selected for brewing in this study that were generated from the 2-row barley genotypes: Copeland, Expedition, Full Pint, Meredith, Metcalfe, and PolarStar (Table 1). All genotypes are widely used for both domestic and craft brewing and are considered interchangeable to produce major beer styles. Further, the malt sources included four growing environments (Montana and Oregon, U.S.A.; Alberta and Saskatchewan, Canada), from four regional malt-houses: Rahr Corporation, Malteurop Group, Briess Malt and Ingredients Company, and Cargill, Inc.

2.2. Brewing method

The beer for this study was brewed at the Haas Innovations “Innovation Brewery” in Washington, U.S.A. on a 2.5-hectoliter system. This recipe was designed to be malt-forward for the purposes of this study, and was consistent for the malt quality specifications of the sample set (Supplementary Table 1). Hop addition was limited to 8 international bittering units (IBU), strictly as a bittering component for the beer; ale yeast (*S. cerevisiae*) was sourced from New Belgium Brewing; water was minimally treated with Calcium (100 ppm), SO₄ (65 ppm), and Cl (95 ppm) to achieve a pH of ~5.4 for mash-in. Yeast pitch-rate was 10⁶/mL/P + 10⁶ and fermentation temperature was held at 20 °C until completion, after which the beers were bottled and pressurized under CO₂. The specifications set for the purposes of this study were intended to reduce variation and provide normalization of quality among the resulting beers (e.g. taste, aroma, appearance, mouthfeel). Beers were produced in three rounds of two beers during each round.

2.3. Malt and beer metabolite extraction

Malt was milled using a Thomas® Wiley Mini-Mill (Thomas Scientific, Swedesboro, NJ), using a #40 mesh sieve. The mill was thoroughly cleaned using 70% ethanol before and between malt sources to reduce metabolite contamination among samples. 100 mg of ground malt was placed into 1 mL vials and extracted using an MTBE:MeOH biphasic method (Supplementary Methods Files 1–2). Bottled beer was equilibrated to room temperature and 50 µL aliquots were taken for non-volatile metabolite analysis, dispersed into separate, labeled autosampler vials with 100 µL inserts for direct-injection and stored in –20 °C until further analysis (Supplementary Methods Files 1–4).

2.4. Metabolomics and ionomics via UHPLC-, HILIC-, and GC-, and ICP-MS

Malt and beer extracts were analyzed using non-targeted metabolomics on multiple mass spectrometry (MS) systems: reverse-phase ultra-performance liquid chromatography-mass spectrometry (UHPLC) (Supplementary Methods File 1), hydrophilic interaction liquid chromatography-mass spectrometry (HILIC) (Supplementary Methods File 2), and beer volatiles were profiled using headspace solid-phase micro extraction gas chromatography–mass spectrometry (HS-SPME/GC–MS) based on methods as previously described (Broeckling, Heuberger, & Prenni, 2013; Broeckling, Heuberger, Prince, Ingelsson, & Prenni, 2013; Heuberger, Broeckling et al., 2014; Heuberger et al., 2012; Turner et al., 2016) and are fully available in Supplementary Methods File 3. Metals were profiled with an ionomics method based on inductively coupled plasma MS (ICP-MS) (Supplementary Methods File 4).

Metabolomics data processing was performed as previously described (Broeckling et al., 2014; Gowda et al., 2014; Smith, & al., e., 2006). Briefly, each sample resulted in a matrix of molecular features (defined by retention time and mass (*m/z*)) generated using XCMS software in R v.3.2.4 (Smith, et al. 2006). Samples were normalized to the total ion current (TIC) and the relative abundance (quantity) of each molecular feature was determined by the mean area of the chromatographic peak among replicate injections (*n* = 2). Mass spectra were deconvoluted using the RamClust algorithm (Broeckling et al., 2014). Compounds were annotated based on retention time and spectral matching using RamSearch software (Broeckling et al., 2014) that included in-house libraries of authentic standards, as well as to external libraries NIST v.14 (<http://www.nist.gov>), Metlin (Tautenhahn et al., 2012; Zhu et al., 2013), the Human Metabolome Database (Wishart et al., 2013), and the Golm Metabolome Database (Hummel et al., 2013).

2.5. Sensory analysis

Sensory analysis of the six beers was performed by a trained panel at New Belgium Brewing Company (Fort Collins, CO, USA). Beer was stored at room temperature during weeks zero through 4 and stored at 4 °C during weeks 4 through 8. Beer was evaluated for sensory traits at 0, 4, and 8 weeks after bottling. Beer was evaluated using a modified Quantitative Descriptive Analysis® (QDA) for 45 sensory traits. Standard QDA was modified to fit the parameters for quantification of beer sensory. For each evaluation event, 16 expert panelists (9 men and 7 women, age 25–55, median age 32 years) were chosen based on their experience and training in several sensory profiling events. (Stone & Sidel, 2004a, 2004b, 2004c). Each panelist was trained according to ASTM (American Society for Testing and Materials) (ASTM, 2017; Chambers & Wolf, 1996) and received 3 samples in a random order (the samples were tasted in a different order during each session) in two sets and approximately 5 min within each set. Panelists were provided palate-cleansing water and crackers between samples as described by Zhao et al. (2018). Panelists, trained in descriptive sensory, were first required to establish robust qualitative sensory descriptors for each sample (e.g. toasty, roasty, malty, umami) during a blind tasting session, based on prior experience with beer sensory. During a subsequent session, the panelists were instructed to determine a perceived quantity for each trait that encompasses taste, aroma, mouthfeel, and appearance. Attributes were then pared down to a clarified lexicon, a final list of the most commonly mentioned attributes (reported in this study). The panelists also utilized reference standards for ratings when available (e.g. Cheerios™, Honeycomb Cereal™, bread crumb). Panelists were monitored weekly to ensure performance and accuracy as described by Dong et al. (2014). Flavors which were considered “off” were given standards, if available. In further tests, panelists were required to score each sample on a scale from 0 to 5 for the 45 attributes possible: “0” indicated that the attribute was undetectable and “5” indicated the highest intensity of the attribute. These attributes were defined and rated during different phases of tasting for aroma, mouthfeel, and taste. Scale use and alignment was agreed upon in a series of three 1-h trainings where standards were determined, and wort and beer samples were evaluated as a group to determine the intensity of each attribute. Scaling of some attributes was voluntary; the attributes that were mandatory to scale were: nutty, corn ships, bread crust, grainy – Grape Nuts™, Honeycomb Cereal™, grassy, hay, watermelon rind/cucumber, green apple/ethyl hexanoate, pear, white grape, honey, caramel, perfume, honeysuckle, sulfidic, sulfidic, musty, Play-doh™ (Barr, 2018) (Supplementary Table 2). Sensory scoring tests were repeated at 0, 4, and 8 weeks in triplicate as described by Stone & Sidel (2004a, 2004b, 2004c).

2.6. Statistics

Statistical models were used on the annotated malt and beer metabolites and the 45 beer sensory traits. Principal Component Analysis (PCA) was conducted on mean-centering and unit variance-scaled metabolite and sensory data using SIMCA software v.14.1 (Sartorius Stedim Biotech, Umea, Sweden) (Umetrics, 2012). Metabolite abundances were compared using two-way ANOVA, via the *aov* function in the R statistical environment v. 3.2.4 for malt and beer genotypes (each of six) and growing location (U.S. or Canada) with a *p* threshold of 0.05. False discovery rate adjustment was performed on the ANOVA *p*-values using the Benjamini-Hochberg algorithm (Benjamini & Hochberg, 1995). O2PLS models were conducted in SIMCA v. 14.1 on the scaled and centered data sets for sensory traits (*y*) and annotated metabolites (*x*). O2PLS models were cross-validated using a leave one out approach (Bouhaddani et al., 2016; Bylesjo, Eriksson, Kusano, Moritz, & Trygg, 2007; Umetrics, 2016). Data were *z*-transformed and resulting *z*-scores were used to create heatmaps utilizing hierarchical clustering and Spearman's rank correlation (*r_s*) methods, and also variable line plots.

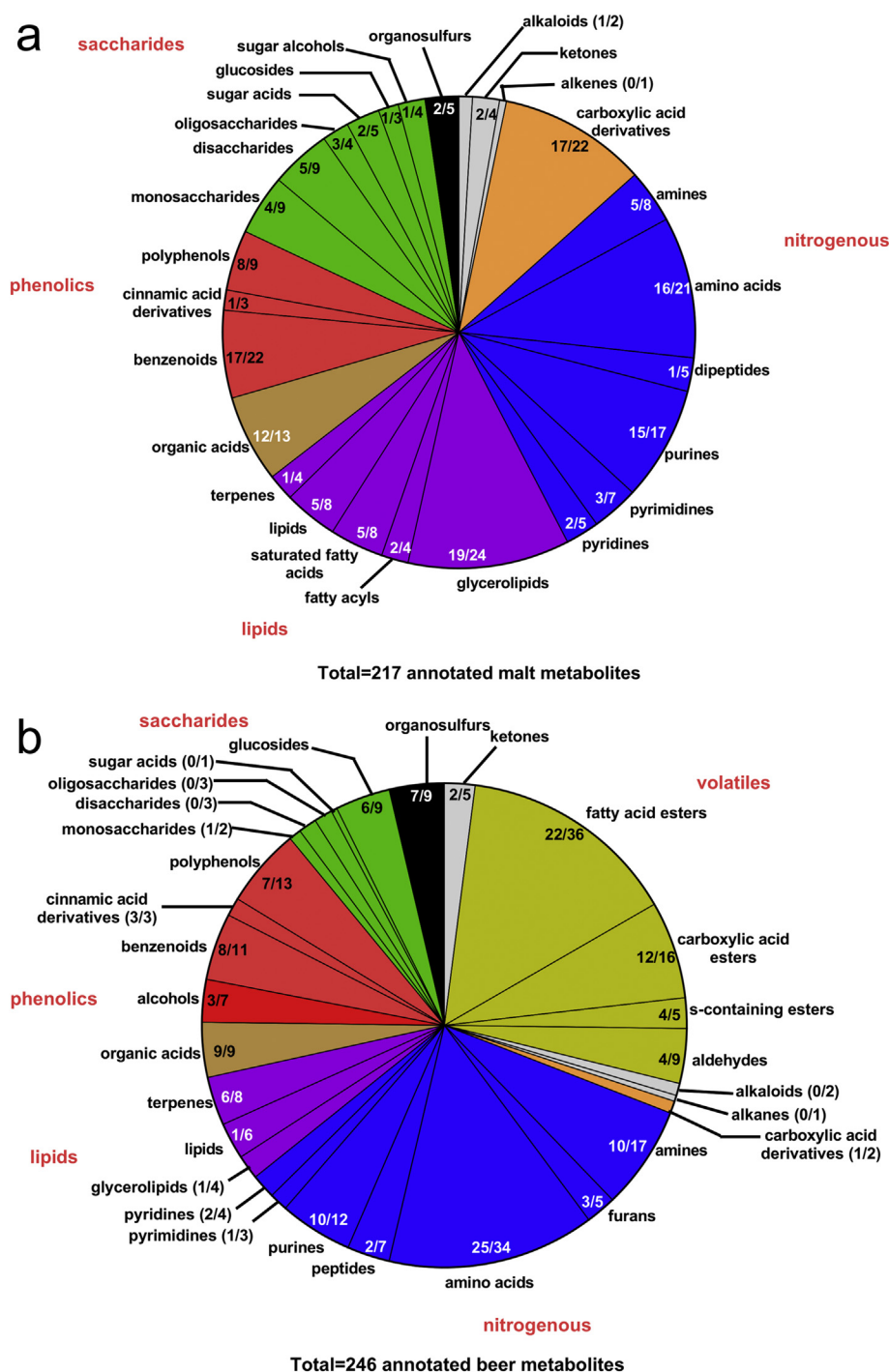


Fig. 1. Annotated malt and beer metabolites and the corresponding chemical class. A total of 217 and 246 metabolites were annotated in (a) malt and (b) beer extracts. Pie charts display metabolites by broader chemical class (red text) and sub-class (black text), and number of metabolites that varied within sub-class are indicated as a value/total metabolite within the sub-class (ANOVA, FDR adjusted $p \leq 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Heatmaps with hierarchical clustering were generated in R using the *ggplots2*, *Reshape2*, and *stats* packages with the *heatmap.2*, *melt*, and *hclust* functions (Warnes et al., 2015; Wickham, 2010, 2014).

3. Results

3.1. Metabolomics analysis of malt and beer revealed chemical variation attributed to malt source

Malt metabolite and metal variation was determined using a non-

targeted metabolomics and ionomics workflow, and profiles were established using three mass spectrometry (MS) platforms: non-volatile metabolites via UHPLC-MS and HILIC-MS, and ICP-MS for metals. The UHPLC-MS and HILIC-MS platforms detected 2492 and 2550 metabolites, respectively. Of these detected compounds in malt, 217 were annotated as metabolites. The six malts were evaluated for metabolite variation using analysis of variance (ANOVA). The analysis revealed 150 of the 217 annotated compounds (69.1%) varied among genotype (ANOVA, FDR adjusted $p \leq 0.05$). Metabolites within all chemical classes varied among malts (Fig. 1a, Supplementary Table 3,

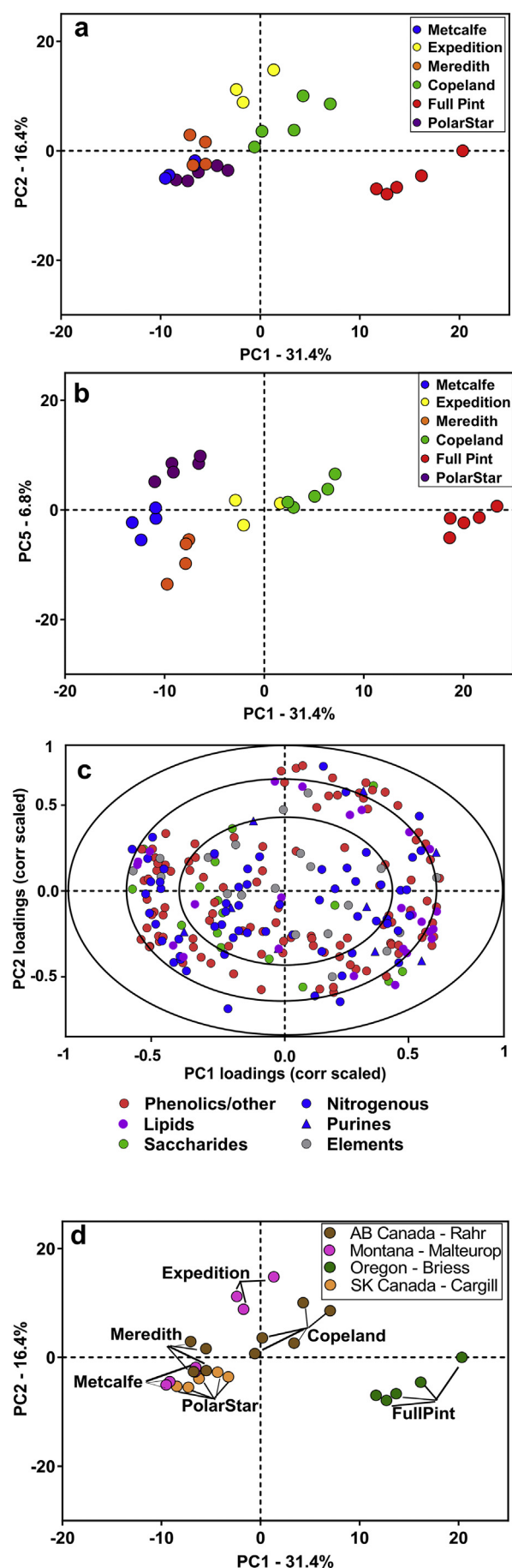
“Annotated metabolites detected in malt and beer”). Further, 46% of all metabolites varied between locations (ANOVA, FDR adjusted $p \leq 0.05$, 2311 out of the 5042 metabolites). A total of 20 metals were analyzed in malt (Supplementary Table 4), including copper (Cu), iron (Fe), calcium (Ca), and sulfur (S), which are important for yeast nutrition and brewing. Metals that varied among malts included Se, Na, Mg, K, Ca, Mn, Fe, Zn, Sr, and Ba, however this variation did not extend to difference among the six beers (ANOVA, $p \leq 0.05$ for all metals).

The beer metabolite profile was chemically similar to malt extracts with the additional detection of volatiles (via HS-SPME/GC-MS metabolomics). The four MS platforms detected 1659 (UHPLC-MS), 2057 (HILIC-MS), and 852 compounds (HS-SPME/GC-MS), and 21 metals (ICP-MS), for which 246 metabolites were annotated (108 non-volatiles, 138 volatiles). ANOVA revealed 150 of the 246 annotated metabolites (60.9%) varied among genotype (FDR adjusted $p \leq 0.05$). Metabolites within all chemical classes varied (Fig. 1b, Supplementary Table 3). This variation in beer metabolites included non-volatiles (nitrogenous compounds/sugars) and volatiles (esters/aldehydes/ketones) according to the genotype and location grown (Canada or U.S.) and revealed that 27% of all metabolites varied between Canada and the U.S. (ANOVA, FDR adjusted $p \leq 0.05$, 1267 out of the 4568 metabolites).

PCA was performed to evaluate variation among the malt metabolite profiles and was independently performed for metabolites and metals. The data set included the 217 annotated malt metabolites or 20 metals. For metabolites, a total of 5 PCs were generated that explained the variation among the six malts (Fig. 2a). PC1 and PC2 explained 31.4% and 16.4% of the variation (respectively) and separated three genotypes Full Pint, Copeland, and Expedition from a cluster of the other three genotypes (Meredith, Metcalfe and PolarStar) (Fig. 2a). Further separation of the varieties is explained by PC5 (6.5%) (Fig. 2b). The PC1 and PC2 loadings (Fig. 2c) indicated some trends in metabolite classes that influence the variation among the six malts. Specifically, lipids were generally higher in Full Pint compared to the other five genotypes. Other metabolite classes, such as amino acids were generally equally distributed in the loadings plot, indicating none of the six genotypes exhibit a major trend in being higher or lower in any one chemical class. Further, the PCA was evaluated to understand if metabolite variation could be attributed to maltster. The PC1 and PC2 scores plot was colored according to each of the four maltsters (Fig. 2d). The data revealed that for the two maltsters that were replicated (Rahr and Malteurop), the respective genotypes were on different locations on the PC scores plot. This indicates that our experiment detected no clear impact of maltster on the metabolite profiles.

Beer metabolite variation was also evaluated by PCA. The data set included the 246 annotated beer metabolites and 21 metals. A total of 6 principal components were generated, of which PC1 and PC2 explained 41.4% of the variation within the volatile metabolite set (Fig. 3a). PC1 and PC2 explained 45.1% of the variation within the non-volatile metabolite set (Fig. 3b). The PC scores plot demonstrated each of the six beers had a distinct metabolite profile (Fig. 3a and b).

For the 20 malt metals, PCA resulted in 5 principal components. The PC scores plot of PC1 and PC2 (42.7% and 14.3% of the variation) separated most of the sources (Supplementary Fig. 1a). Of the six malts, Full Pint and PolarStar had the most distinct metals profiles. The PC loadings plot indicates that Full Pint was higher in trace metals such as copper (Cu), zinc (Zn), and manganese (Mn) (Supplementary Fig. 1b). However the variation in malt metals did not extend to metals variation in beer, as indicated by the lack of separation on the PCA scores plot (Supplementary Fig. 1c). There were 5 principal components generated that explained variation among the six beers (Supplementary Fig. 1c). PC1 explained 42.1% of the variation and PC2 explained 15.1% of the variation. Full Pint was separated from the other sources, largely due to Mn, Zn, sulfur (S), and iron (Fe) (Supplementary Fig. 1d).



(caption on next page)

Fig. 2. Principal component analysis (PCA) of malt metabolites of the six malt sources. PCA was performed on the 217 annotated malt metabolites for the six malts. (a) PC scores plot for PC1 and PC2 of six malts. (b) PC scores plot for PC1 and PC5 provides additional separation among the malt sources. (c) Correlation scaled loadings plot for PC1 and PC2 of the six malts, colored according to chemical class. (d) PC scores plot (PC1 vs PC2) colored according to each of the four maltsters. Analyses were based on $n = 3$ –5 extraction replicates per source.

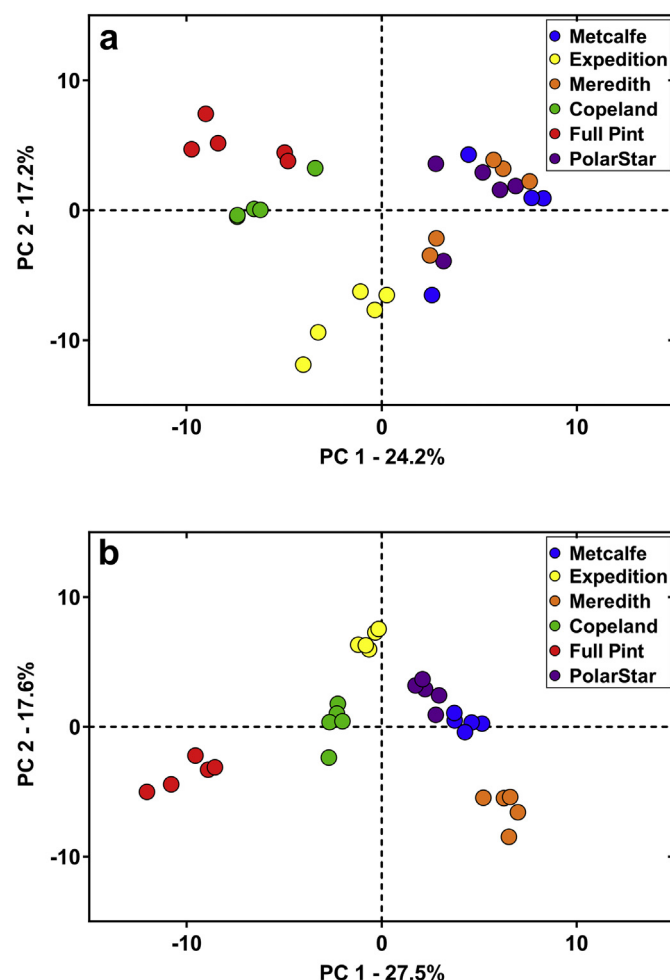


Fig. 3. Principal component analysis (PCA) of beer metabolites of the six malt sources. PCA was performed on the annotated metabolites for the six beers. PCA scores plot were produced based analysis of (a) the 138 volatile metabolites and (b) 108 non-volatile metabolites. Analyses were based on $n = 3$ –5 extraction replicates per genotype.

3.2. Sensory differences were observed in beer after two months of storage

The metabolomics data support that malt source influenced the metabolite profiles of the resulting beers. Next, a sensory experiment was performed to determine if the malt and beer metabolite variation influenced flavor qualities of the beer. The brewing recipe used for the six malts was designed to utilize an industrial scale system based on the concern that brewing on pico- or micro-levels would be difficult to replicate and compare malts for an influence on flavor (Supplementary Table 1). The recipe is relatively malt-forward with low hop levels and yeast with low flavor-producing esters, designed to evaluate malt flavor. The general consensus of the panel was that distinct flavors in beers were discernable after 2 months of storage. Sensory analysis of the six beers was conducted using modified quantitative descriptive analysis® (QDA) of 45 traits that encompass taste, aroma, and mouthfeel. PCA was conducted on the scores generated by the QDA panel for

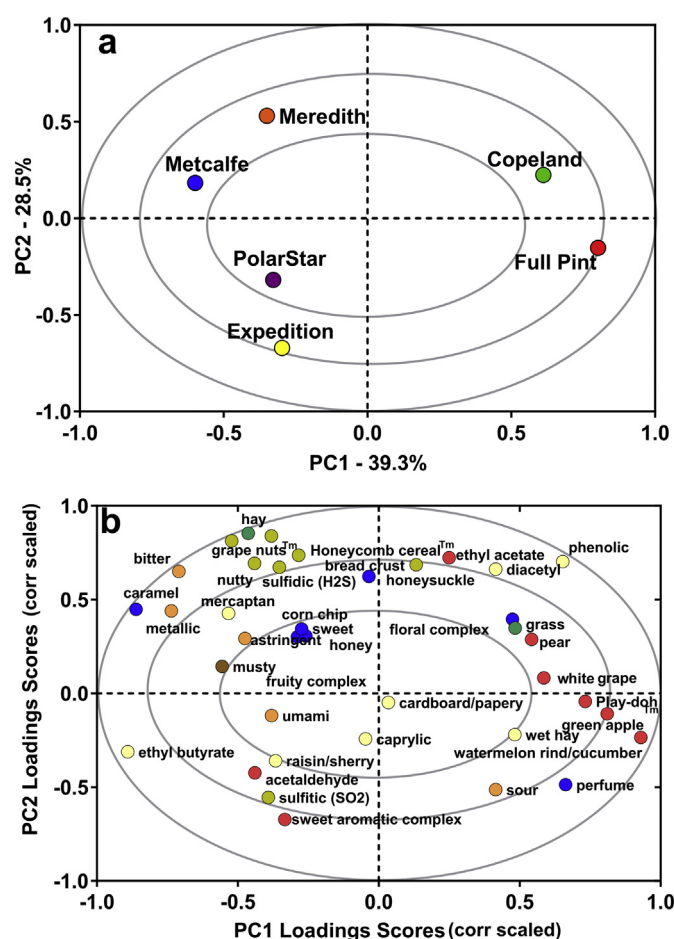


Fig. 4. Principal component analysis (PCA) of sensory data from the six beers after 2 months of storage. PCA was performed on data for 45 sensory traits quantified for the six beers. (a) PC scores plot based on panel consensus data with (b) corresponding correlation-scaled loadings plot. Loadings were colored according to broad sensory trait.

the sensory traits on beer at 2 months of storage and supports differences in overall sensory profiles of the six beers (Fig. 4a). At Month 2, 3 PCs were generated explaining 83% of the variation. The loadings plots indicates that sensory profile differences were largely influenced by 17 of the 45 sensory traits (Fig. 4b, correlation scaled loadings $> |0.75|$).

3.3. Previously characterized beer flavor metabolites did not co-vary

The annotated malt and beer metabolites were searched in flavor reference databases to assign known descriptors for the influenced on sensory traits (“flavor-class”, Supplementary Table 3) (ASBC, 2015; Kim et al., 2016; Wishart et al., 2013). The beer PCA loadings were then replotted by color-coding each metabolite’s flavor class for cardboard/papery, umami, corn chip/sulfidic, sulfidic, and fruity (Supplementary Fig. 2a–f). The loadings plots revealed a scattered distribution of metabolites and an overall lack of co-variation based on flavor-class. However, individual metabolites were identified whose flavor-class was consistent with the flavor reported by the sensory panel (Supplementary Fig. 2a–f).

3.4. Beer sensory profiles are associated with metabolite variation in beer and malt

The metabolite and sensory data were integrated using two analyses: (i) two-way orthogonal projection to latent structures (O2PLS) and (ii) Spearman’s correlation analysis. O2PLS is a method that

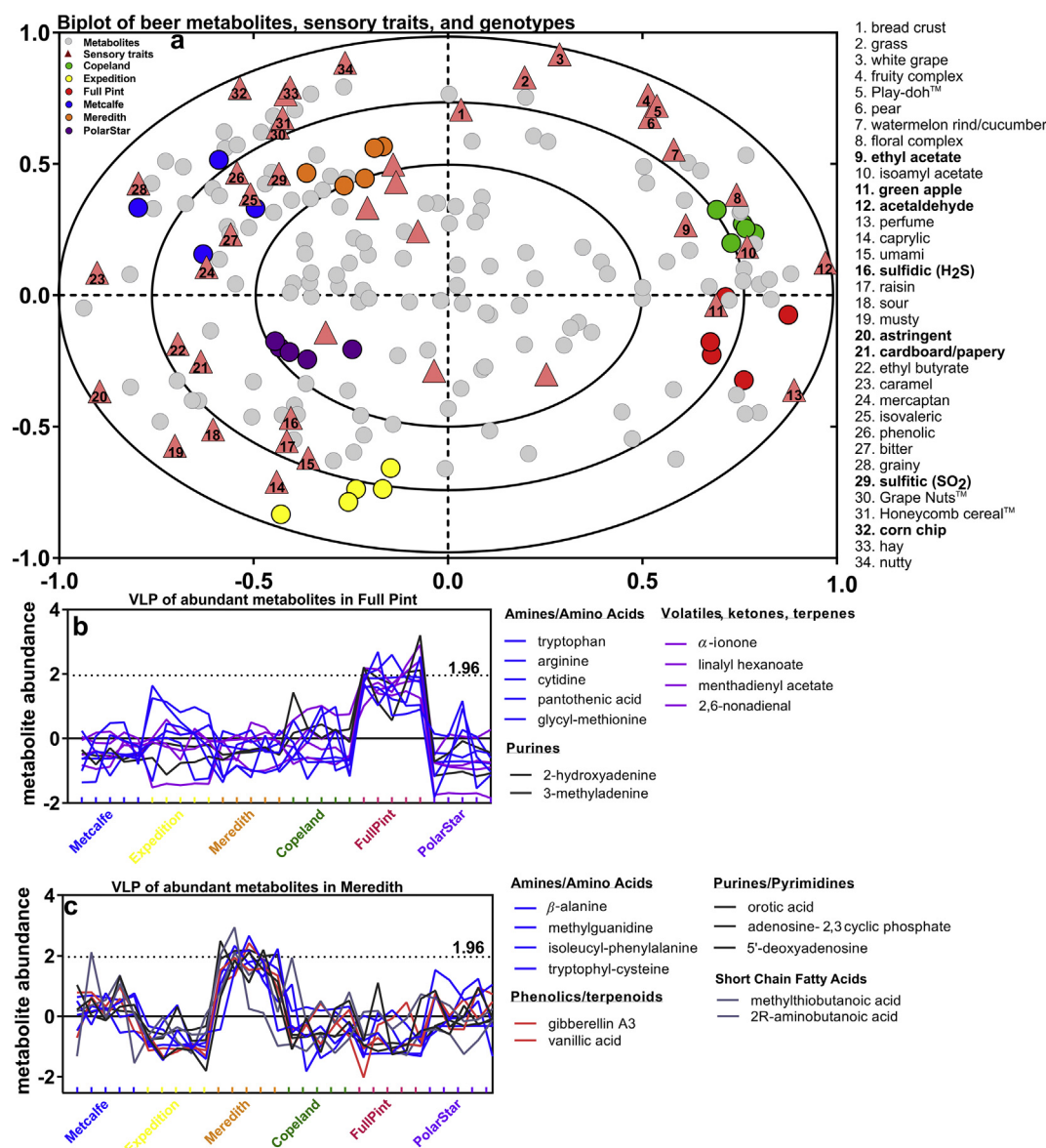


Fig. 5. Multivariate association of beer metabolites with beer sensory traits. The association between beer metabolites and beer sensory was evaluated with two-way orthogonal partial least squares (O2PLS) and performed on 246 metabolites, 21 metals, and 45 sensory traits (a) O2PLS scores and loadings plot of the metabolites/metals (gray circles), sensory traits (triangles) and malt sources (colored circles). Sensory traits associated specifically with Meredith or Full Pint are denoted in bold text. Numbers in triangles correspond to sensory traits. Variable line plots (VLP) display normalized abundance variation of metabolites that are higher in a specific malt source for (b) Full Pint and (c) Meredith. z of 1.96 indicates 95% confidence that a metabolite is higher in a sample compared to the other five malt sources.

integrates two multivariate data sets based on co-variation in the system, and has been used in metabolomics studies (Bouhaddani et al., 2016; Bylesjo et al., 2007; Heuberger, Broeckling, et al., 2014). O2PLS was conducted on the malt, beer, and sensory traits and indicated several metabolites that co-vary with sensory. Full Pint and Copeland were associated with *fruity*, *watermelon rind*, *ethyl acetate*, *isoamyl acetate*, and *acetaldehyde/green apple* sensory traits (Fig. 5a). Meredith and Metcalfe were associated with the *corn chip*, *sulfidic (H₂S)*, and *Honeycomb cereal* sensory traits, and PolarStar and Expedition were associated with *umami* and *cardboard* flavors (Fig. 5a). The O2PLS model was able to predict > 50% of the variance in 25 of the 45 sensory traits using metabolite data ($Q^2 > 0.4$, $p < 0.05$) (Supplementary Fig. 3a). An analysis of loadings was used to determine beer metabolites associated with a distinct flavor property of the beer (Tables 2 and 3). These trends were visualized as z -scaled variable line plots (VLP) for the loadings associated with Full Pint and Meredith, indicating individual metabolites that were more abundant in each of the beers (Fig. 5a, b,

and c, Tables 2–5). Full Pint beer (*fruity*) contained several nitrogenous compounds (notably cytidine, tryptophan, arginine, the purines 2-hydroxyadenine and 3-methyladenine), and terpenes consistent with fruit-like or perfume flavor (e.g. linalyl hexanoate, α-ionone) (Fig. 5b, Table 2). Meredith (*corn chip*) was also high in several amino acids and terpenes, but notably sulfur-containing compounds such as 2-methylthiobutanoic acid and tryptophyl-cysteine (Fig. 5c, Table 3).

The beer metabolite-sensory associations were further evaluated using Spearman's rank correlation combined with hierarchical clustering and heat mapping (Fig. 6a). The full correlation matrix was reduced to show “contributors” to variation, in this case 46 metabolites that met the correlation threshold and were nitrogenous. The metabolites that contributed to the unique flavor profile of Meredith were several amino acids and terpenes, but there were more sulfur-containing compounds associated with this source (i.e. 2-methylthiobutanoic acid and tryptophyl-cysteine). This indicates that the unique flavor profiles from each source are the combination of many metabolites

Table 2
Beer metabolites associated with flavor in Full Pint.

Chemical class	Metabolite name	Structure	Platform ^a	Reported sensory	Beer flavor-class	Malt flavor-class
Alcohols	Phenethyl alcohol	C8H10O	c	Alcohol, flowery, honey-like, roses, sweet	Wet hay/acetalddehyde	
Aldehydes	2,6-Nonadienal	C9H14O	c	Cucumber, watermelon	Watermelon rind	
Aldehydes	Methyl benzoate	C8H8O2	c	Fruity, herbal, floral	Acetalddehyde	
Amines	1-Acetonylpyrrolidine	C6H13NO	a		Play-doh [™]	
Amines	5-Methylcytosine	C5H7N3O	a		Perfume/fruity/green apple/Play-doh [™]	Acetalddehyde
Amines	Cytidine	C9H13N3O5	b	Umami, bitter	Watermelon/ethyl acetate/Play-doh [™]	Play-doh [™]
Amines	<i>n</i> -Acetyl-methionine	C7H13NO3S	a/b	Sulfurous, umami		
Amines	Nicotinamide riboside	C11H15N2O5	a		Fruity/perfume/Play-doh [™] /green apple	Caramel/nutty/umami/bitter
Amines	Thiamine	C10H14N2O5	b	Sweet, nutty	Green apple	
Amines	Thiamine	C12H17ClN4OS	b			
Amino acids	Acetyl glycine	C4H7NO3	b	Fruity	Fruity/perfume/ethyl acetate/green apple	
Amino acids	β -Alanine	C3H7NO2	b		Wet hay/acetalddehyde	
Benzenoids	Benzaldehyde	C7H6O	c	Bitter almond, cherry stone, almond	Floral/grass	Floral/grass/ethyl acetate/watermelon rind
Benzenoids	β -Ionone (2,4-di-tert-butylphenol)	C14H22O	c	Raspberry, citrus, woodlands, violet, kettle hop aroma, artificial raspberry, strawberry, floral, berry	Wet hay/acetalddehyde	
Esters	Isopentyl hexanoate	C11H22O2	c	Fruity, solvent, perfumed, tropical fruits	Fruity/ethyl acetate/white grape/watermelon/green apple	
Esters	Methyl decanoate/methyl caprate	C11H22O2	c	Sweet, coconut, fruity	Bread crust/fruity	Ethyl acetate/green apple
Esters	Methyl tetradecanoate	C15H30O2	c	Perfume, herbal, petals	Perfume/ethyl acetate/fruity/watermelon/Play-doh [™]	Perfume/ethyl acetate/green apple
Esters	Nonyl phenylacetate	C17H26O2	c	Fruity, fruit, soapy, tropical, tea-like	Fruity/ethyl acetate	Metallic
Fatty acid esters	Butyric acid	C5H10O2	c	Rotten, fruit punch	Watermelon rind/cucumber/fruity/white grape	
Ketones	2,2-Dimethyl-1,3-cyclohexanedione	C8H12O	c	Sweet, caramel, maple	Acetalddehyde	
Lipids	Butanoic acid	C5H10O2	b	Buttery, rancid, cheesy	Play-doh [™]	
Lipids	Caprylic acid	C8H16O2	c	Goaty, vegetable oil, wet dog	Acetalddehyde	Green apple/ethyl acetate
Lipids	<i>n</i> -Methyl- α -aminoisobutyric acid	C5H11NO2	a		Perfume/fruity/green apple/Play-doh [™] /ethyl acetate	Ethyl butyrate/bread crust
Organic acids	Citraconic acid	C5H6O4	b	Citric	Grassy/white grape/floral/ethyl acetate	Green apple/ethyl acetate
Phenolics	Apigenin-6-C-glucoside	C21H19O10-	a	Grassy, hoppy	Fruity/perfume/ethyl acetate/green apple/isoamyl acetate	Bitter/astringent, caramel
Saccharides	Tagatose	C6H12O6	a/b	Sweet	Play-doh [™]	
Saccharides	Maltopentaose	C36H52O26	a	Sweet	Perfume/ethyl acetate/fruity/watermelon/Play-doh [™]	
Saccharides	Oligosaccharide		a		Perfume/ethyl acetate/fruity/watermelon/Play-doh [™]	
Saccharides	Stachyose	C24H42O21	b		Green apple/fruity	Astringent/grainy
Saccharides	Sucrose	C12H22O11	b	Sweet	Sweet/honey/ethyl butyrate	Green apple/ethyl acetate
Saccharides	Trehalose	C12H22O11	a/b	Sweet	Play-doh [™] /green apple/fruity/ethyl acetate	Green apple/ethyl acetate
Saccharides	Turanose	C12H22O11	a/b	Sweet	Ethyl acetate/perfume/fruity	Ethyl acetate/perfume/fruity
Terpenes	Linyl hexanoate	C16H28O2	a	Pineapple, warm, animal, fruity, pear, green	Fruity, green apple	

^a Platform – a denotes UHPLC-MS; b denotes HILIC/LC-MS; c denotes HS-SPME/GC-MS; d denotes if the metabolite was found in malt, e denotes if the metabolite was found in beer, sensory associated with beer at Month 2, according to the O2PLS beer model.

Table 3
Beer metabolites associated with flavor in Meredith.

Chemical class	Metabolite name	Structure	Platform ^a	Sensory detected	Beer flavor-class	Malt flavor-class
Alcohols	Furazan-3-ol, 4-amino	C2H3N3O2	c	aromatic, roasty, nutty	com chip/Honeycomb Cereal™/nutty	Play-doh™
Aldehydes	Cinnamaldehyde	C10H10O	a	spicy, bitter, phenolic	honeycomb/sweet/sweet-aromatic/honey/Honeycomb Cereal™	
Aldehydes	Hydroxymethylfurfural	C6 H6O3	a	stale, papery/vegetal/bready, caramel	com chip/Honeycomb Cereal™/nutty	sulfidic/cardboard/sulfidic
Aldehydes	Methional	C4H10O5S	c	umami	umami/grainy/sulfurous	
Alkaloids	N-Methyltryptamine	C11H14N2	a		umami/astringent/cardboard	
Amines	2-Methylpyrrole	C5H7N	c	sulfuric, bitter	com chip/Honeycomb Cereal™	
Amines	4-Methylpyridine	C6H7N	c	roasted, nutty, cocoa, peanut	phenolic/sulfidic	wet hay
Amines	5-Hydroxytryptophol	C10H11NO2	a	old almonds	musty/sulfidic/sulfidic	pear/grass/fruity
Amines	Dithiouracil	C4H4N2S2	c	roasted corn, toasted bread	bread crust	sulfidic/sulfidic/wet hay
Amines	4-Dimethylaminopyridine	C7H9N5	b		umami/sulfidic/sulfidic	sulfidic
Amines	Ethyl dimethylcarbamate	C5H11NO2	c		com chip/Honeycomb Cereal™/nutty/sulfidic	
Amines	Methionine sulfoxide	C5H11NO3S	a		sulfidic/sulfidic/wet hay	
Amines	Methylamine	C6H13NO	c	vegetal, grape, carrot, cabbage	sulfidic	wet hay
Amines	Ornithine	C5H12N2O2	b		com chip/honeysuckle/Honeycomb Cereal™	sulfidic/cardboard/sulfidic
Amines	Pyroglutamic acid	C5H7NO3	b	soapy, astringent	umami	corn chips
Amines	Sarcosine, N-(3-phenylpropionyl)	C17H25NO3	c		com chip/Honeycomb Cereal™/metallic	
Amines	Orotic acid	C5H4N2O4	b	matches, sulfurous	nutty/sulfidic/sulfidic/Honeycomb Cereal™/grainy	
Amines	Niacin	C6H5NO2	b/a	sour, metallic	com chip	metallic
Amino acids	Arginine	C6H14N4O2	a	sweet, bitter, astringent	bread crust	grainy-grape nut cereal
Amino acids	Asparagine	C4H8N2O3	b/a	savory	bread crust	bread crust
Amino acids	Biotin	C16H28N4O4S	a		Honeycomb Cereal™/nutty/com chip	pear/isoamyl acetate
Amino acids	Isoleucine	C6H13NO2	a	bitter-sweet, astringent	sulfidic/sulfidic/grainy/musty/isovaleric/umami	
Amino acids	Lysine	C6H14N2O2	a	lemony	honey comb/nutty/com chip/grainy	
Amino acids	Methionine	C5H11NO2S	b/a		isovaleric/com chip/umami/	umami
Amino acids	Tryptophan	C11H12N2O2	b/a	bitter-sweet, methanol-like	sulfidic/sulfidic/corn chip/ethyl butyrate	grainy/astringent/mercaptan
Amino acids	Pantothenic acid	C9H17NO5	b/a	astringent, salty	bitter/astringent/sour/isovaleric/ethyl butyrate	sulfidic/sulfidic
Amino acids	α-Aminobutyrate	C4H9	a	sweet-bitter, sour, umami	com chip/Honeycomb Cereal™/honeysuckle	
Benzenoids	para-aminobenzoic acid	C7H7NO2	b	hay-like	wet hay	sulfidic/sulfidic/phenolic
Esters	Allyl-alanine	C6H11NO3S	c		ethyl butyrate/corn chip/isovaleric	mercaptan
Esters	Isobutyl 3-methyl-2-butenate;	C9H16O2	c	green, spicy, mint	nutty/sulfidic/sulfidic/phenolic	phenolic
Esters	Isobutyl caprylate	C12H24O2	c	swiss cheese, winery, fatty, sweet	ethyl butyrate/caprylic/isovaleric/corn chip	mercaptan
Esters	Methyl 2-(methylthio)butyrate	C6H12O2S	c	musty, onion, sulfurous	isovaleric/mercaptan/sulfidic/sulfidic	sulfidic/sulfidic
Esters	Methyl 4-(methylthio)butyrate	C6H12O2S	a		sulfidic	
Esters	Methylthio-2-(propanoyloxy)propanoate	C7H12O3S	c	sulfuric ester	honey comb/nutty/corn chip/grainy/sulfidic/sulfidic	
Furans	5-Methyl hexanoate	C7H14O2 S	c	cooked vegetable, sulfuric, soapy	bread crust	
Furans	2,4-Dimethylphenyl formamide	C6H8O4	c	roasty, nutty	com chip/Honeycomb Cereal™/nutty	metallic
Furans	Furan, 2-nonadecanoyl	C23H40O2	c	roasty, nutty	com chip/Honeycomb Cereal™/nutty	
Ketones	2-Propionylfuran	C7H8O2	c	roasty, nutty	com chip/Honeycomb Cereal™/nutty/sulfidic	
Lipids	2-Hydroxy-4-(methylthio)butyric acid	C5H10O3S	a	rotten	com chip/grainy/sulfidic/nutty/umami	
Organic acids	Fumaric acid	C4H4O4	c	sour	sour/sulfidic/phenolic	sulfidic/sulfidic/sour
Organosulfurs	4-(methylthio)-1-butanamine	C5H13NS	c	cabbage, garlic, potato, sulfuric, vegetal	sulfidic/sulfidic/mercaptan	sulfidic/phenolic
Organosulfurs	Benzothiazole	C7H5NS	c	rubbery, sulfuric, gasoline	mercaptan/astringent	wet hay/metallic/sulfidic
Organosulfurs	Methyl propyl disulfide	C4H10S2	c	garlic, burnt rubber	Honeycomb Cereal™	corn chips/honey comb
Peptides	Glycyl-methionine	C7H14N2O3S	a		bread crust	
Peptides	Isoleucyl-phenylalanine	C15H25N2O3	a		sulfidic/corn chip	
Peptides	Prolyl-arginine	C11H21N5O3	a		com chip/nutty/sulfidic	
Peptides	Prolyl-cysteine	C8H14N2O3S	a		sulfidic	
Peptides	Tryptophyl-cysteine	C14H17N3O3S	a		sulfidic/ethyl butyrate/corn chip/grainy	
Phenolics	Vanillic acid	C8H8O4	b		nutty/sulfidic/sulfidic/Honeycomb Cereal™/grainy	acetaldehyde
Phenolics	Xanthohumol	C21H22O5	b	bitter		phenolic
Purines	5-Doxysadenosine	C10H13N5O3	a	umami	sulfidic/sulfidic/caprylic/umami/caramel/phenolic/acetaldehyde	

(continued on next page)

Table 3 (continued)

Chemical class	Metabolite name	Structure	Platform ^a	Sensory detected	Beer flavor-class	Malt flavor-class
Purines	5-Methylthioadenosine	C11H15N5O3S	b/a	umami, sulfurous	nutty/corn chip/grainy/sulfidic/sulfidic	phenolic/sulfidic
Purines	Adenine	C5H5N5	b/a		corn chip	Play-doh™/nutty/grainy
Purines	Adenosine	C10H13N5O4	a	bitter, umami, corn chip	sulfidic/grainy/corn chip	Play-doh™/watermelon rind
Purines	Inosine	C10H12N4O5	b	meaty, savory	bread crust	bread crust
Purines	Purine	C5H4N4	b		watermelon rind/cucumber/fruity/white grape/isoamyl	phenolic/nutty/sulfidic/bitter
Saccharides	Isomaltose	C12H22O11	a		acetate	hay/corn chips
Terpenes	Linalyl hexanoate	C16H28O2	a	pineapple, warm, animal, fruity, pear, green	Honeycomb Cereal™/nutty/corn chip/grainy	
Terpenes	α-Ionone	C13H20O	a	floral, sweet, woody, tropical, fruity, raspberry, violet	bread crust	

^a Platform – a denotes UHPLC-MS; b denotes HILIC/LC-MS; c denotes HS-SPME/GC-MS; d denotes if the metabolite was found in malt, e denotes if the metabolite was found in beer, sensory associated with beer at Month 2, according to O2PLS beer model.

found in beer. Purines and pyrimidines found in malt and beer which are subject to Maillard reactions have been associated with the *corn chip* flavor in beer (Harding, Nursten, & Wren, 1977). Other nitrogenous compounds (amines, amino acids, purines) in beer were associated with flavor traits. For example, 5-methylthioadenosine (a purine nucleoside) was negatively correlated ($r_s < 0.5$) with most fruity traits (e.g. *green apple* and *watermelon rind*), but was positively correlated with typical staling traits (e.g. *cardboard*, *bitter*, *astringent*) (Fig. 6a). Other negatively correlated ($r_s < 0.71$) amino acids and dipeptides in beer were also dominant in Meredith, such as biocytin and the flavor traits associated with it (corn chip, bread crust, sulfidic, sulfidic). Interestingly, biocytin did not display a positive relationship with the typical staling traits (cardboard, umami, sulfidic), and therefore Meredith's high abundance, yet negative correlation of sensory traits to biocytin warrants further investigation. Thymine (a DNA nucleobase) and cytidine (a nucleoside), co-varying metabolites, were associated with the fruitier flavors of Full Pint. Proline, an amino acid which is not readily absorbed by yeast during fermentation, was noted as more abundant in Meredith and Full Pint (Fig. 6).

O2PLS analysis was further performed on the malt metabolites and beer sensory, and associations were observed among malt and beer quality. Separation among sources was explained by sensory traits and the metabolites associated with them in the O2PLS overview biplot (Fig. 7a, Tables 4–5). As with the beer model, O2PLS established relationships between Full Pint malt and sensory (*fruity*, *sweet*, *acetaldehyde*, *watermelon rind*), and Meredith malt with its beer sensory traits (*corn chip*, *sulfidic*, and *Honeycomb cereal™*). A total of 16 out of 45 sensory traits in the malt model had cumulative prediction rates of 50% or greater (Supplementary Fig. 3), indicating which sensory traits could be reliably predicted based on malt metabolite data. An analysis of the O2PLS loadings for Full Pint revealed that some phenolics, organic acids and sugars (Fig. 7b, Table 4), as well as some malt lipids may be associated with sensory (Fig. 7c, Table 4). In Meredith malt, sulfur-containing amines, amino acids and sugars were associated with *Honeycomb cereal™* and *corn chip* flavor (Fig. 7d, Table 5). Taken together, the malt O2PLS analysis support that each source had a unique flavor profile based on the integration of several classes of malt metabolites (Tables 4 and 5).

Spearman's correlation analysis of malt metabolite and beer sensory was performed independently for two chemical classes due to the known contribution to brewing and beer flavor: nitrogenous compounds (important for yeast fermentation) and lipids (important for mouthfeel, flavor, foam, and oxidation reactions), and for metabolites indicated as important to sensory based on the O2PLS analysis. Several nitrogenous malt metabolites were correlated to beer sensory traits (Fig. 8a). Clustering of the flavors *green apple*, *isoamyl acetate*, *Play-doh™*, and *fruity* were positively correlated ($r_s > 0.71$, $p \leq 0.05$) with isoleucyl-proline (amino acid), and 2-ketohexanoic acid, an oxo-keto-acid. 2-ketohexanoic acid is metabolized by yeast during fermentation and is involved in the formation of fusel alcohols from aldehydes (Pires, 2015; Pires, Teixeira, Branyik, & Vicente, 2014) (Fig. 8b). The *corn chip*, *caramel*, and *Grape Nuts™* flavor traits had relationships with, for example, γ -glutamyl-methionine (a peptide), ornithine (an amine, the result of arginine catabolism by lactic acid bacteria in the mashing stage of brewing) and purines. Tyramine (an amino acid) and 5-MTA were correlated to *bitter*, *musty*, *umami*, *astringent* and *Grape Nuts™* flavors in malt. Several malt metabolites were associated with beer sensory traits at Month 2, however integration of these metabolites is key to the production of flavors (Fig. 8). For lipids, variation among malt source was overserved among lipid classes (Supplementary Fig. 4). The triglyceride class and saturated fats such as palmitic and stearic acid, was associated with the *fruity* and *watermelon rind* sensory traits. Full Pint was generally higher in the saturated fat class of lipids (Supplementary Fig. 4a and b). Phosphatidylcholines and glycerophosphocholines were correlated with *corn chip*, *Grape Nuts™*, and *sulfidic* sensory traits. These classes of lipids are recognized to have an effect on beer flavor, foam

Table 4

Malt metabolites associated with flavor in Full Pint beer.

Chemical class	Metabolite name	Structure	Platform ^a	Reported sensory	Beer flavor-class	Malt flavor-class
Alkenes	1-Pentadecene	C15H30	a			Perfume/acetaldehyde/ green apple
Amines	Cytidine	C9H13N3O5	b	Umami, bitter	Watermelon/ethyl acetate/ Play-doh™	Acetaldehyde
Amines	Glutathione	C10H17N3O6S	b			Green apple/Play-doh™
Amines	n-(1-deoxy-fructosyl)methionine	C11H21NO7S	a			Acetaldehyde/perfume
Amines	n-Acetyl-methionine	C7H13NO3S	a/b	Sulfurous, umami		Play-doh™
Amines	n-Acetyl-phenylalanine	C11H13NO3	b			Acetaldehyde
Amines	n-methyl-D-aspartate	C5H9NO4	b	Sour, umami		Acetaldehyde
Amines	n-Oleoyl-alanine	C21H39NO3	a			Perfume/acetaldehyde/ green apple
Amines	Pyrrolidonecarboxylic acid	C5H7NO3	a/b			Acetaldehyde/green apple
Amines	Ribothymidine	C10H14N2O6	b			Play-doh™
Amino acids	Allothreonine	C4H9NO3	b	Sweet, bitter, astringent		Play-doh™/watermelon rind
Amino acids	Valine	C5H11NO2	a/b	Sweet, bitter, astringent		Fruity/Play-doh™
Benzenoids	4-Aminosalicylic acid	C7H7NO3	b	Fruity-astringent		Green apple/ethyl acetate/ perfume
Benzenoids	Benzaldehyde	C7H6O	c	Bitter almond, cherry stone, almond	Floral/grass	Floral/grass/ethyl acetate/ watermelon rind
Benzenoids	Creosotinic acid	C8H8O3	b			Green apple/ethyl acetate
Esters	Carboximide acid	C18H37NO2	a			Green apple/ethyl acetate
Lipids	2,6-Dimethyl-7-octene-1,6-diol 8-O glucoside	C16H30O7	a	Anise-like, fennel		Green apple/ethyl acetate
Lipids	2-Amino-octadecanoic acid	C18H37NO2	a			Acetaldehyde
Lipids	2-Phenylbutyric acid	C10H12O2	b			Acetaldehyde
Lipids	Alchornoic acid	C20H36O3	a			Perfume/acetaldehyde/ green apple
Lipids	Aanandamide (17:1)	C19H37NO2	b			Perfume/fruity/Play-doh™
Lipids	Caprylic acid	C8H16O2	a	Goaty, fatty acid, vegetable oil, wet dog	Acetaldehyde	Green apple/ethyl acetate
Lipids	Citraconic acid	C5H6O4	b	Citric		Green apple/ethyl acetate
Lipids	Deoxyglycylglycine	C26H43NO5	b			Perfume/Play-doh™/fruity
Lipids	Elaidic acid	C18H34O2	b			Perfume/green apple/ethyl acetate
Lipids	GPE (p-18:0/20:4)	C5H14NO6P	b			Green apple/ethyl acetate
Lipids	Heptanoic acid	C7H14O2	b	Rancid		Green apple/perfume/ acetaldehyde
Lipids	LysoPC (18:2)	C26H50NO7P	a			Green apple/ethyl acetate
Lipids	LysoPC (18:3)	C26H48NO7P	a			Green apple/ethyl acetate
Lipids	Nervonic acid	C24H46O2	b			Perfume
Lipids	n-Methyl-α-aminoisobutyric acid	C5H11NO2	a		Perfume/fruity/greenapple/ Play-doh™/ethyl acetate	Ethyl butyrate/bread crust
Lipids	Oleic acid	C18H34O2	b			Fruity
Lipids	PC (18:4)	C26H46NO7P	a			Green apple/ethyl acetate
Lipids	PC (20:0)	C36H72NO8P	b			Green apple/ethyl acetate
Lipids	PC (32:0)	C40H80NO7P	b			Green apple/ethyl acetate
Lipids	PC (34:0)	C34H70NO6P	b			Play-doh/watermelon rind/ fruity
Lipids	PC (16:0/18:1)	C42H82NO8P	b			Green apple/ethyl acetate
Lipids	PC (18:4)	C40H80NO8P	a			Green apple/acetaldehyde
Lipids	PC (O-14:0)		a			Green apple/ethyl acetate
Lipids	PE (40:1)	C48H94NO8P	a			Green apple/ethyl acetate
Lipids	PG-NMe (32:0)	C38H76NO8P	b			Green apple/ethyl acetate
Lipids	PGP (32:0)	C38H75O10P	a			Green apple/ethyl acetate
Lipids	PGP (36:4)	C42H76O13P2	a			Green apple/ethyl acetate
Lipids	PI (46:0)	C52H104NO12P	a			Green apple/ethyl acetate
Lipids	Turanose	C12H22O11	a/b	Sweet	Ethyl acetate/perfume/fruity	Ethyl acetate/perfume/ fruity
Organic acids	Ketoisocaproate	C6H10O3	b	Sweet, fruity		Metallic/acetaldehyde
Organic acids	l-2-Hydroxyglutaric acid	C5H8O5	b			Ethyl acetate/green apple
Organic acids	Malic acid	C4H6O5	b	Sweet-sour		Green apple/ethyl acetate
Organic acids	Oxoadipic acid	C6H8O5	b			Acetaldehyde/perfume
Organooxygens	N-Acetyl-Neuraminic Acid	C11H19NO9	b			Perfume/acetaldehyde/ green apple
Phenolics	3-Ethylphenol	C8H10O	c			Perfume/acetaldehyde/ green apple
Phenolics	Isoferulic acid	C10H10O4	a/b	Clove, spicy, fruity		Perfume/Play-doh™/ acetaldehyde/green apple
Phenolics	n-[4-(cyanomethyl)phenyl]-5- (phenoxymethyl)furan-2-carboxamide	C20H16N2O2	c			Green apple/ethyl acetate
Phenolics	Shikimate	C7H10O5	b			Green apple
Purines	1-Methyladenosine	C11H15N5O4	b	Milk-like, salty		Play-doh™/perfume

(continued on next page)

Table 4 (continued)

Chemical class	Metabolite name	Structure	Platform ^a	Reported sensory	Beer flavor-class	Malt flavor-class
Purines	2-Phenylami	C16H18N6O4	b			Green apple/ethyl acetate
Saccharides	Arabitol	C5H12O5	b			Green apple/fruity
Saccharides	Fructose	C6H12O6	b	Sweet		Green apple/ethyl acetate
Saccharides	Galactitol	C6H14O6	b	Sweet		Green apple/ethyl acetate
Saccharides	Stachyose	C24H42O21	b		Green apple/fruity	Astringent/grainy
Saccharides	Sucrose	C12H22O11	b		Sweet/honey/ethylbutyrate	Green apple/ethyl acetate
Saccharides	Tagatose	C6H12O6	a/b	Sweet	Fruity/perfume/ethylacetate/ green apple/isoamylacetate	Bitter/astringent, caramel
Saccharides	Trehalose	C12H22O11	a/b	o-Glycosyl compounds	Playdoh/greenapple/fruity/ ethyl acetate	Green apple/ethyl acetate

^a Platform – a denotes UHPLC-MS; b denotes HILIC/LC-MS; c denotes HS-SPME/GC-MS; sensory associated with beer at Month 2, according to O2PLS beer model.

retention, and mouthfeel (Cuzzolino et al., 2015; Cuzzolino & Degner, 2016). The lipid content and composition varied in each source in the study (Supplementary Fig. 4b), and the abundances of malt lipids also varied among lipid class.

4. Discussion

The main findings of this study include that (i) six commercial malt sources (defined by barley genotype and growing location) with similar malting quality contained significant metabolite variation, spanning many metabolite classes, (ii) this metabolite variation is related to metabolite variation in the beer and (iii) this impacted aging-related flavor chemistry pathways within the beer, and sensory differences were discernable by a trained panel.

Metabolomics and ionomics were performed with UHPLC-MS, HILIC-MS, ICP-MS, and HS-SPME/GC-MS in combination with a modified quantitative descriptive analysis to profile the changes between malt and beer, the metabolites involved, and the predicted co-varying metabolites that could be used to describe flavor and flavor stability in beer. Beer flavor is largely driven by the interaction and variation of specific combinations of malt metabolites with macro-level chemistry (i.e. total extract, total protein) during fermentation. The abundances of most metabolites varied among the six commercial malt sources. Several previous studies have reported a contribution of barley genotype to malt metabolite variation (Herb et al., 2017; Heuberger, Broeckling et al., 2014). For this study approximately 23% and 61% of the annotated metabolites varied among the six malts and beers (respectively), and this was distributed among up to 10 different chemical classes (Fig. 1). The malt data included 217 metabolites, and several with known effects on fermentation and flavor including lipids, purines, and amines. The beer metabolite data included 246 metabolites, several with known effects on flavor such as purines, amines, phenolics, alkaloids, esters and terpenes. Here, two of the six malts/beers were further evaluated in detail for associations between malt chemistry, beer chemistry, and effects on flavor: Full Pint (unique *fruity* flavor) and Meredith (unique *corn chip* flavor).

4.1. Effect of metabolites on beer flavor from Full Pint malt

Full Pint is a variety released by Oregon State University with enhanced agronomic and disease resistance properties (Verhoeven et al., 2011). A previous study crossed Full Pint barley with another variety (Oregon Promise), and an integrated genetic and sensory analysis found that Full Pint had higher *malt*, *sweet*, *toasted*, and *toffee* flavor compared to Golden Promise (Herb et al., 2017). Here, Full Pint malt produced beer with fruitier flavor descriptors compared to the five other malts. The differences in the data may be attributed to several factors, such as growing environment, malting, brewing (recipe and scale of production), aging of the beer, and that both studies are limited to analyses that compare samples within an experimental system. Thus, understanding how malt source influences the chemistry of the beer is

foundational to begin to form a consensus from independent studies on beer flavor.

In our study, Full Pint beer was differentiated by the sensory descriptors *fruity*, *ethyl acetate*, and *pear* at Month 2, all fruit-based traits. The Full Pint beer chemistry revealed several nitrogenous compounds such as cytidine, 5-methylcytosine, adenine, and thymine (all forms of DNA nucleobases) which are important factors to consider, as these may contribute downstream to FAN composition (Lekkas, Stewart, & al., e., 2005; Schwarz & Li, 2010). Beer is a pyrimidine-rich beverage, for example, the cytidine nucleotide is excreted by yeast both early in fermentation and also during aging, which suggests a possible contribution to flavor in beer. (Figs. 5 and 6) (Bokulich & Bamforth, 2013; Gupta et al., 2010; Schwarz & Li, 2010). Cytidine has been previously observed in beer and was high in Full Pint (Fig. 6b) (Briggs, 2004; D. Briggs, 1998; Hough, Briggs, Stevens, & Young, 1982; Spevacek, Benson, Bamforth, & Slupsky, 2016). Cytidine is dependent upon the amount of sulfites, a natural product of fermentation, in the beer and may play a role in the creation of *fruity* flavor profiles in beer over time, possibly due to interactions with hop products and the subsequent breakdown of compounds due to aging (Briggs, 2004; Hughey et al., 2015; Spevacek et al., 2016). Further, Full Pint beer was higher in tryptophan and arginine, and these amino acids can influence beer flavor during aging, interacting with volatile compounds, purines, or alkaloids. Full Pint beer was also higher in purines 2-hydroxyadenine and 3-methyladenine (Fig. 5b). During brewing, purines are absorbed by yeast, but are not held by them and release purines back into the beer (Hughey et al., 2015; Johnson et al., 2013; L. Li et al., 2015). Because of this, purine content is high in beer and has been shown to interfere with or influence hop-related flavors, such as myrcene, phenolics, and amino acids (Briggs, 2004; Hughey et al., 2015; Spevacek et al., 2016). The purine nucleotide class is recognized to have an influence on flavor in beer when concentration levels are high (Briggs, 2004; D. Briggs, 1998; Liu, 2015; Liu, Dong, Yin, Li, & Gu, 2014).

Two classes of volatile metabolites were detected that would influence Full Pint's fruit-like flavor: carbonyls (aldehydes and ketones) and monoterpenes. Full Pint beer was highest in α -ionone (Figs. 5 and 6), a volatile ketone/sequiterpene associated with *floral*, *pear*, and *watermelon rind* attributes (Dong et al., 2015). Interestingly α -ionone, may be synthesized from the beer terpene citral and acetone (acetone as a product of free fatty acid degradation that can occur during aging), or from carotenoid degradation (Bravi, Marconi, Sileoni, & Perretti, 2017; Cuzzolino et al., 2015; Cuzzolino & Degner, 2016; Horak et al., 2009; Strating & Eerde, 1973). Therefore, in the Full Pint beer, terpene and lipid degradation pathways may interact to form new terpenes that were described by the panel *fruity*, *watermelon rind*, and *ethyl acetate* in Full Pint (Figs. 5b and 6b) (De Keukeleire, 2000). Further, in most beer, the monoterpenes linalool and geraniol (*floral*), and limonene and α -terpineol (*citrus*) are commonly associated with fruit-like flavor (Dong et al., 2015; Inui et al., 2013; Schönberger & Kostecky, 2011). Monoterpenes were present, but not abundant in Full Pint beer in our study (Tables 2–5).

Table 5
Malt metabolites associated with flavor in Meredith beer.

Chemical class	Metabolite name	Structure	Platform ^a	Reported sensory	Beer flavor-class	Malt flavor-class
Alkaloids	<i>n</i> -Methyltryptamine	C11H14N2	a	Umami	Umami/astringent/cardboard	Sulfidic/cardboard/sulfidic
Amines	3- <i>o</i> -methylidopa	C10H13NO4	b	Bitter, lit match		Corn chip/grainy/sulfidic/phenolic
Amines	5-Hydroxytryptophol	C10H11NO2	a	Old almonds	Caprylic/musty/sulfidic/sulfidic	Pear/grass/fruity
Amines	Glutamine	C5H11NO2	b	Fruity, vegetal, umami, savory		Bread crust/corn chip
Amines	maleic acid	C4H4O4	b			Bread crust
Amines	<i>n</i> -(1-deoxy-1-fructosyl)leucine	C12H23NO7	a			Sweet/bread crust/Honeycomb Cereal TM
Amines	<i>n</i> -(1-deoxy-1-fructosyl)isoleucine	C12H23NO7	a			sulfidic/sulfidic/sour
Amines	<i>n</i> -Acetylglutamic acid	C7H11NO5	b	Sulfurous, umami		Sulfidic/sulfidic/phenolic/cardboard
Amines	Ornithine	C5H11NO2	b			Sulfidic/cardboard/sulfidic
Amines	<i>o</i> -Tyrosine	C9H11NO3	b			Corn chips/Honeycomb Cereal TM
Amines	Pyroglutamic acid	C5H7NO3	b	Soapy, astringent	Umami	Mercaptan/corn chips
Amines	Uracil	C4H4N2O2	b			Sulfidic/sulfidic
Amines	Niacin	C6H5NO2	a/b	Sour, Metallic	Corn chip	Metallic
Amino acids	Alanine	C3H7NO2	b	Sweet		Phenolic/sulfidic
Amino acids	Aminooctanoic acid	C8H17NO2	b			Bread crust
Amino acids	Arginine	C6H11NO4	a	Sweet, bitter, astringent	Bread crust	Grainy-Grape Nuts Cereal TM
Amino acids	Asparagine	C4H8N2O3	a/b	Savory	Bread crust	Bread crust
Amino acids	Biocytin	C16H28N4O4S	a		Honeycomb Cereal TM /nutty/corn chip	Pear/isoamyl acetate
Amino acids	Diaminopimelic acid	C7H14N2O4	b			Bread crust
Amino acids	Methionine	C5H11NO2S	a/b		Isovaleric/corn chip/umami/	Umami
Amino acids	Tryptophan	C11H12N2O2	a/b	Bitter-sweet	Sulfidic/sulfidic/corn chip/ethyl butyrate	Grainy/astringent/mercaptan
Amino acids	Tyrosine	C9H11NO3	a	Cheddar cheese		Sulfidic/sulfidic
Amino acids	Tyrosine	C9H11NO3	b	Vegetal, savory		Bread crust/Honeycomb Cereal TM
Amino acids	Pantothenic acid	C9H17NO5	a/b	Astringent, salty	Bitter/astringent/sour/isovaleric/ethyl butyrate	Sulfidic/sulfidic
Benzenoids	3-Phenoxypropionic acid	C9H10O3	b			Phenolic/sulfidic
Benzenoids	Para-aminobenzoic acid	C7H7NO2	b	Hay-like	Wet hay	Sulfidic/sulfidic/phenolic
Lipids	Hydroxyisocaproic acid	C6H12O3	b	Goaty, sulfurous		Sulfidic/sulfidic/phenolic/cardboard
Lipids	PA (34:2)	C37H69NO8P	b			Nutty/sulfidic
Lipids	TG (50:5)	C52H98O6	a	Sour	Sour/sulfidic/phenolic	Sulfidic/sulfidic/sour
Organic acids	Fumaric acid	C4H4O4	c			Nutty
Organic acids	Galactonate	C6H12O7	b	Acidic		Nutty
Organic acids	Lactic acid	C3H6O3	b	Sweet-sour, tart		Green apple/ethyl acetate
Organic acids	Malic acid	C4H6O5	b			Corn chips/hay/mercaptan
Peptides	Glutamyl methionine	C10H16N3O6	a	Toasty, roasty, malty		Sweet/bread crust
Peptides	Glycylproline	C7H12N2O3	a/b		Nutty/sulfidic/sulfidic/Honeycomb Cereal TM /grainy	Acetaldehyde
Phenolics	Vanillic acid	C8H8O4	b			Sulfidic/ethyl butyrate/sulfidic
Phenolics	Rutin	C27H30O16	b			Phenolic/sulfidic
Purines	5-Methylthioadenosine	C11H15N5O3S	a/b	Umami, sulfurous	Nutty/corn chip/grainy/sulfidic/sulfidic	
Purines	Adenine	C5H5N5	a/b		Corn chip	Phenolic/sulfidic
Purines	Adenosine	C10H13N5O4	a	Bitter, umami, corn chip		Play-doh TM /nutty/grainy
Purines	Guanosine	C10H13N5O5	b	Umami, grain	nutty/sulfidic/grainy/corn chip	Play-doh TM /watermelon rind
Purines	Inosine	C10H12N4O5	b	Meaty, savory		Astringent/sulfidic/nutty/phenolic
Purines	Purine	C5H4N4	b		Bread crust	Bread crust
Saccharides	Alpha-sophorose	C12H22O11	b			Phenolic/nutty/sulfidic/bitter
Saccharides	Deoxyribose	C5H10O4	a	Sweet		Corn chips/mercaptan
Saccharides	Glucose-1-phosphate	C6H13O9P	b			Honeycomb Cereal TM /corn chip/grainy
Saccharides	Isomaltose	C12H22O11	a			Sulfidic/sulfidic
Saccharides	<i>n</i> -Acetylmannosamine	C8H15NO6	b		Watermelon rind/cucumber/fruity/white grape/isoamyl acetate	Hay/corn chips

^a Platform - a denotes UHPLC-MS; b denotes HILIC/LC-MS; c denotes HS-SPME/GC-MS; sensory associated with beer at Month 2, according to O2PLS beer model.

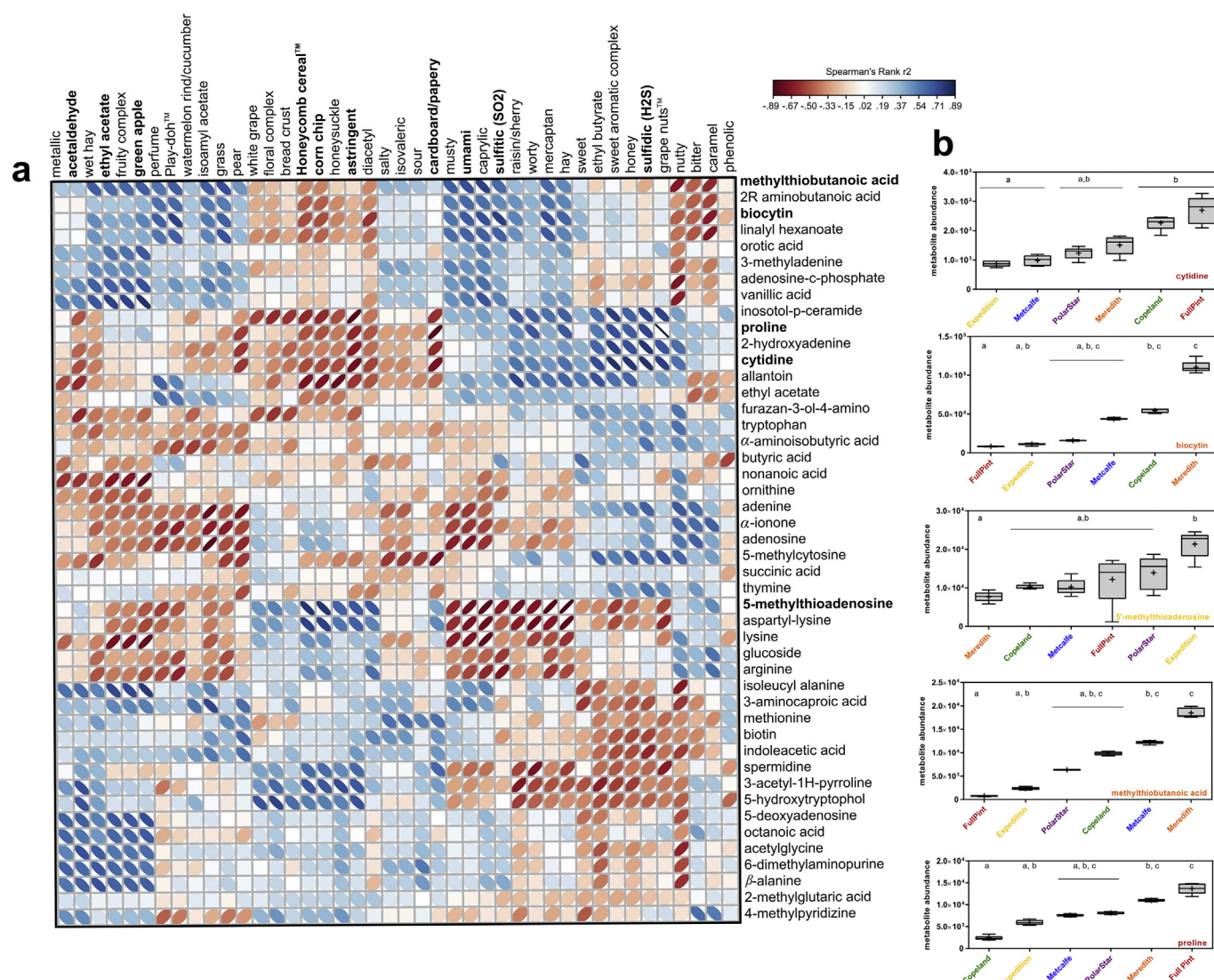


Fig. 6. Univariate association of beer nitrogenous metabolites with beer sensory traits. The association between beer metabolites and beer sensory was determined with Spearman's rank correlation (r_s). (a) Correlation matrix shown as a heat map of r_s values for beer metabolites (46 nitrogenous metabolites, x-axis) and sensory traits (42 sensory traits, y-axis). Colors and the eccentricity of the ellipse indicate strength and direction of the r_s correlation, and the data were arranged using hierarchical clustering. (b) Box plots showing example metabolite variation in the system. Data are presented as the mean metabolite abundance $\pm$ standard error of the mean (S.E.M.) for $n = 3$ –5 extraction replicates per malt source. ANOVA with a Tukey post-hoc ($p \leq 0.05$) was performed on the data in the box plots, and differences are denoted with letters.

Another volatile compound, 2,6-nonadienal, is a di-unsaturated aldehyde, a staling compound and marker of aging in beer and is a derivative of nonanal, which was present, but not abundant in Full Pint beer (Reineccius, 2005). Nonanal is formed via microbially induced lipid oxidation, which, due to the abundance of lipids on Full Pint malt (Fig. 8, Supplementary Fig. 2, Supplementary Table 3), would increase production of 2,6-nonadienal and the subsequent cucumber and watermelon rind sensory traits (Hatanaka, Kajiura, & Harada, 1975). Transamination and decarboxylation of FAN increases the formation of some carbonyls (e.g. 2-methyl- and 3-methyl-butanal, methional, and phenylacetaldehyde from isoleucine, leucine, methionine, and phenylalanine, respectively). The latter amino acids were all present in Full Pint (Figs. 5 and 6) and therefore may also contribute to the creation of fruity characteristics of the beer (Demyttenaere, Tehrani, & De Kimpe, 2002; Nursten, 2005).

Our data also indicate that phenolic metabolites may contribute to fruity flavors. Phenolic/antioxidant activity decreases during the germination phase of malting, but then increases considerably during steeping and kilning (Maillard, Soum, Boivin, & Berset, 1996).

Chlorogenic acid, isoferulic acid, vanillic acid, and p-coumaric acid (phenolic compounds) all have strong antioxidant activity and were prevalent in Full Pint malt (Figs. 7 and 8). Phenolic compounds such as flavan-3-ols (catechins and epicatechins) and hydroxycinnamic acids (isoferulic acid and p-coumaric acid) in malt and beer have a strong impact on the colloidal stability (foam and haze), flavor (astringent, clove), and antioxidant activity (increased shelf-life) of beer (Heuberger et al., 2016; Maillard et al., 1996). Malt phenolics may contribute to the flavor profile of fruity, green apple, and watermelon rind through an interaction with Maillard reaction pathways (Maillard et al., 1996; Nursten, 2005), such as effects on the candy-like maltols. Maillard reaction products during kilning are possible due to the thermal breakdown of carbohydrates during germination, when reducing sugars and amino acids are released. It has been shown that lipoxygenase activity was decreased during kilning due to the increase in phenolic compounds (Maillard et al., 1996; Nursten, 2005).

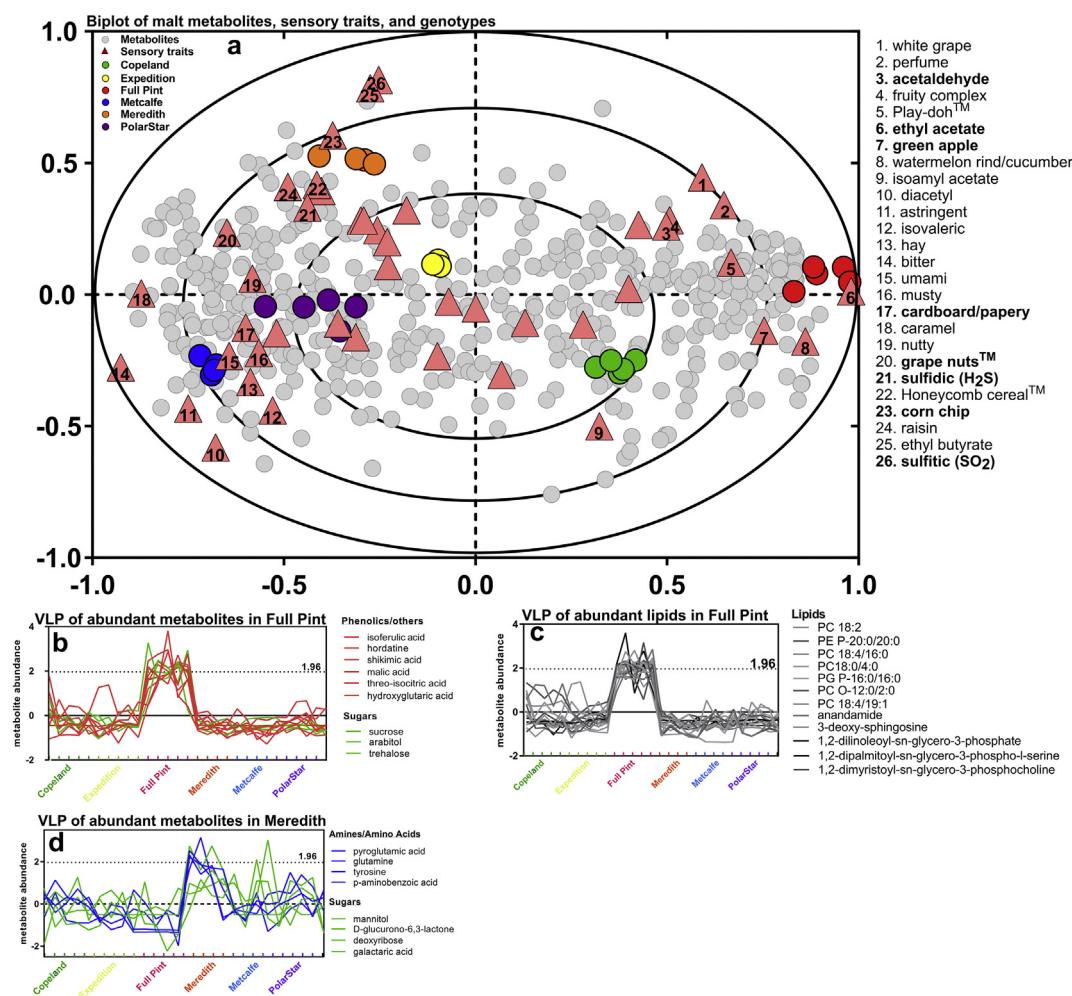


Fig. 7. Multivariate association of malt metabolites with beer sensory traits after 2 months of storage. The association between beer metabolites and beer sensory was evaluated with two-way orthogonal partial least squares (O2PLS) and performed on 217 metabolites, 20 metals, and 45 sensory traits (a) O2PLS overview biplot of all 217 malt metabolites and 20 metals, source, and sensory trait associations showed separation among the six sources and co-variation with sensory traits and metabolites. O2PLS scores and loadings plot of the metabolites/metals (gray circles), sensory traits (triangles) and malt sources (circles). Sensory traits associated specifically with Meredith or Full Pint are denoted as bold text. Numbers in triangles correspond to sensory traits (right of plot). Variable line plots (VLP) graphed the contribution of more highly abundant metabolites contributing to sensory traits associated with each beer source. (b) VLP of malt metabolites (non-lipids) most abundant in Full Pint, as explained by the O2PLS model that includes all metabolites and metals. (c) VLP of malt lipids most abundant in Full Pint. (d) Variable line plot of beer metabolites most abundant in Meredith, as explained by the O2PLS model that includes all metabolites and metals. z of 1.96 approximates 95% confidence that a metabolite is higher in a sample.

4.2. Effect of metabolites on beer flavor from Meredith malt

Meredith is a commonly grown barley variety with overall good malt quality, specifically for its protein and extract composition. Meredith was demonstrated to produce higher soluble protein and FAN, resulting in a relatively darker wort and beer color (MacLeod, Edney, & Izydorczyk, 2011). It is a high-performing, disease-resistant variety that is optimized for enzymatic power and is suited for use in high-adjunct beers, such as light lagers (Mallett, 2014).

In our study, Meredith beer was differentiated by the sensory *corn chip*, *Honeycomb cereal*, and *sulfidic/sulfitic* at Month 2, all savory or sulfuric-based traits. These flavors may be driven by the nitrogenous compounds observed in Meredith beer, such as glutamine and proline (Figs. 5 and 6). The main source of amino acids is found in malt and yeast, as malt contributes FAN, peptides, and polypeptides. Deamination and transamination reactions utilize nitrogenous metabolites precursors to produce several organic acids, aldehydes, esters, and alcohols that influence beer flavor (Gresser, 2009a, 2009b; Taylor & Organ, 2009).

The liberation of FAN in the mash is highly dependent upon

protease activity in the malt. In germinating barley proteinases are responsible for the breakdown of storage proteins into soluble proteins (peptides and amino acids). Proteinase classes do not all show a relationship or correlation to the content of soluble nitrogen that will be available in malt for subsequent mashing (Kihara et al., 2002). It is important to consider the ability of barley (from a breeding perspective) to efficiently degrade grain storage proteins that result in appropriate levels of FAN for brewing. The data indicates several free amino acids that were higher in Meredith, including glutamine, an amino acid related to *umami* (Fig. 7d) (Johnson et al., 2013). Amino acids, purines, and pyrimidines found in malt and beer which are subject to Maillard reactions have been associated with the “corn chip” flavor in beer (as well as the flavor of corn tortillas) (Harding et al., 1977). Reported levels of FAN (as in Table 1) do not currently take into consideration the composition of FAN and the rates at which components are absorbed by ale yeast. As such little is known about the true absorption rate or utilization among varying brewing conditions (Jones & Pierce, 1964; Lekkas et al., 2005). The increase or decrease of specific amino acids can affect the fermentation efficiency and the uptake of other nitrogenous compounds. With an increase in methionine, amino acids

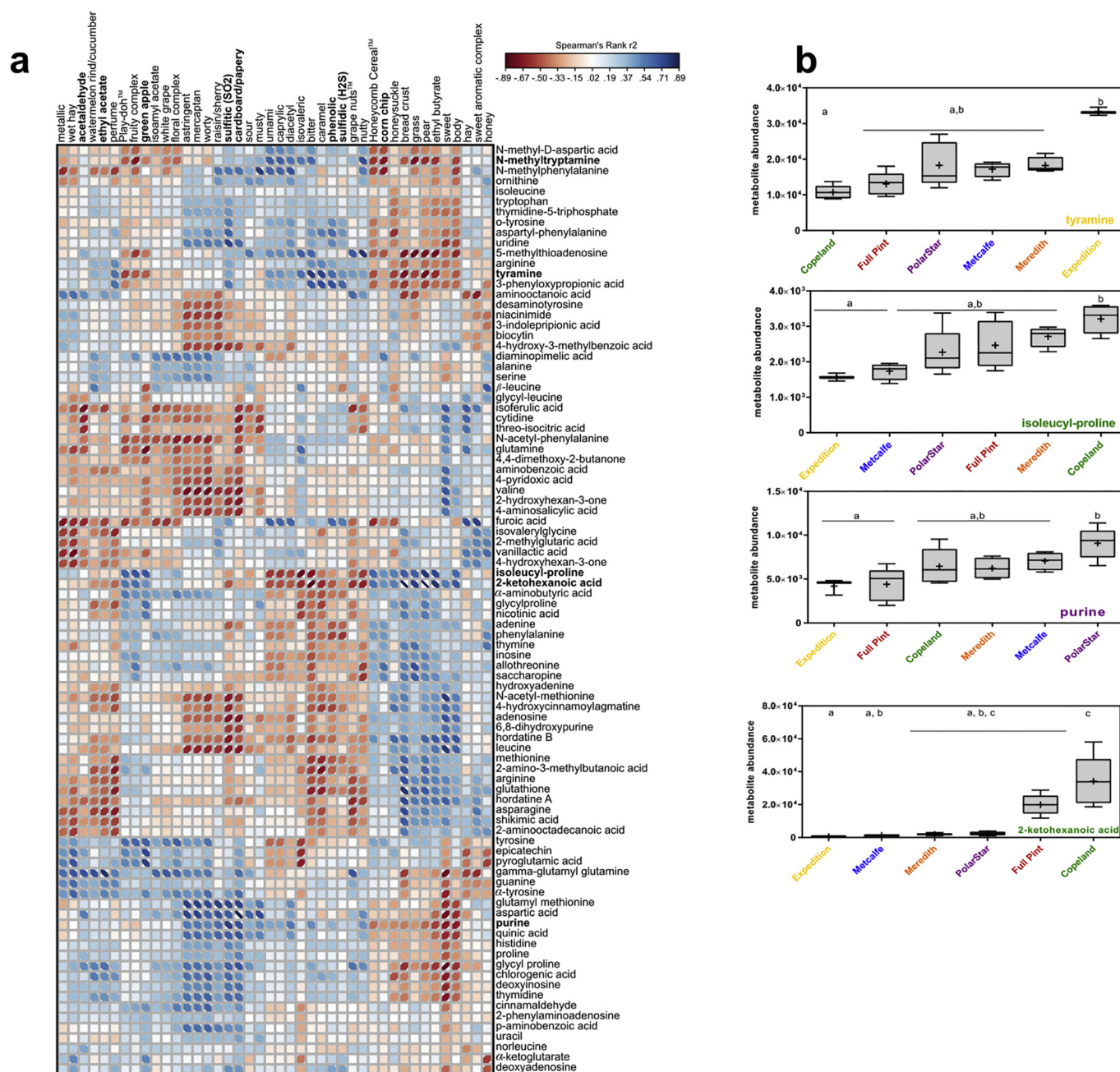


Fig. 8. Univariate association of malt nitrogenous metabolites with beer sensory traits after 2 months of storage. The association between malt metabolites and beer sensory was evaluated with Spearman's rank correlation (a) Correlation plot (heat map) of r_s values for malt metabolites (x-axis) and sensory traits (y-axis). Colors and the eccentricity of the ellipse indicated strength and direction of the r_s correlation, and the data were arranged using hierarchical clustering. (b) Box plots were used to indicate example metabolite variation in the system. Data are presented as the mean metabolite abundance $\pm$ standard error of the mean (S.E.M.) for $n = 3$ –5 extraction replicates per source. ANOVA with a Tukey post-hoc ($p \leq .05$) was performed on the data in the box plots, and differences are denoted with letters.

that are usually absorbed equally as rapidly are not absorbed at all. This underutilization and remainder of amino acids may possibly have a detrimental effect on the fermentation and stability of the beer (Hoki et al., 2013; Jones & Pierce, 1964; Lekkas et al., 2005; Schwarz & Li, 2010; Yu et al., 2014).

Here, Non-volatile and volatile metabolites were associated with *corn chip*, *Honeycomb cereal*, and *sulfidic/sulfitic* sensory traits of Meredith: nitrogenous metabolites (purines and amides), and non-nitrogenous (thio fatty acids). The purine 5-methylthioadenosine (5-MTA), a non-volatile intermediate in the methionine and S-adenosylmethionine (SAM) pathways is a marker for beer aging and flavor stability (Heuberger et al., 2012; Heuberger et al., 2016; Hughey et al.,

2015). Beer with increased 5-MTA has been associated with a *stale* and *corn chip* flavor over time (Heuberger et al., 2016). Baseline levels of 5-MTA were variable among sources in this study, but the trend over time was consistent among beer types in the development of off-flavors and 5-MTA was noted as positively correlated ($r_s > 0.54$) with staling flavors over time (Figs. 5 and 6). Biocytin, a sulfur-containing amide formed from biotin and lysine, (Fig. 6b) was prevalent in Meredith beer. Lysine is readily and quickly absorbed by yeast, however biocytin is not a readily available form of the amino acid (Jones & Pierce, 1964; Pires, 2015; Pires et al., 2014). As biocytin is acted upon by enzymes to make biotin available for metabolism, excess lysine in the beer remains, to break down into components (aldehydes and ketones) that contribute to

flavor instability and savory flavors (Jones & Pierce, 1964; Pires, 2015; Pires et al., 2014). Another interesting compound in Meredith is methylthiobutanoic acid, a short chain thio-fatty acid compound found in beer, generally as a result of yeast desaturase activity, is part of the enzyme complex responsible for fatty acid biosynthesis (Figs. 5c and 6). This compound is found to increase when malt, hop, or yeast quality is poor. It is transformed from amino acids, such as methionine or cysteine, hydrogen sulfide (H₂S) and reactive oxygen into a free sulfur species which is highly reactive and forms other compounds in beer over time with the remaining FAN (Yin et al., 2013).

5. Conclusions

This study investigated the effect of barley source on beer flavor and flavor stability using a non-targeted metabolomics approach. The results support the principles that (i) there are metabolite (small molecule) differences among commercial malt sources, (ii) the malt metabolite differences emerges as distinct metabolite chemistry in the finished beer (iii) the differences in the beer chemistry can impact sensory attributes of beer and (iv) there is therefore a role of malt metabolites in the flavor and flavor stability of the beer. The malt and beer metabolites reported herein can support rapid screening methods to predict effects on flavor and flavor stability, and to better understand how the chemistry of raw ingredients influences beer at the molecular level. Future work is warranted to investigate barley genes and environmental factors that regulate metabolite variation in the malt.

Authors' contributions

AH, CH, HB, LB, DS designed the project. HB, CH, LB, JC executed experiments, HB, LB, CB, JC analyzed the data, and AH, LB, CB, CH, JC, DS, and HB wrote the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodres.2018.07.024>.

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